

IMAGE SHEAR

Computer Vision with Hüseyin Özdemir

Shear with Linear Transformation

Linear Transformation equation for shear

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & \alpha \\ \beta & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

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The diagram illustrates the linear transformation equation for shear. It features three main components: a transformed coordinate vector, a shear matrix, and an input coordinate vector. The transformed coordinate vector, represented as $\begin{bmatrix} x' \\ y' \end{bmatrix}$, is enclosed in a blue box and has a blue arrow pointing to the label "Transformed coordinate vector". The shear matrix, represented as $\begin{bmatrix} 1 & \alpha \\ \beta & 1 \end{bmatrix}$, is enclosed in a red box and has a red arrow pointing to the label "Shear matrix". The input coordinate vector, represented as $\begin{bmatrix} x \\ y \end{bmatrix}$, is enclosed in a blue box and has a blue arrow pointing to the label "Input coordinate vector".

Transformed coordinate vector

Shear matrix

Input coordinate vector

Linear Transformation equation for shear

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & \alpha \\ \beta & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

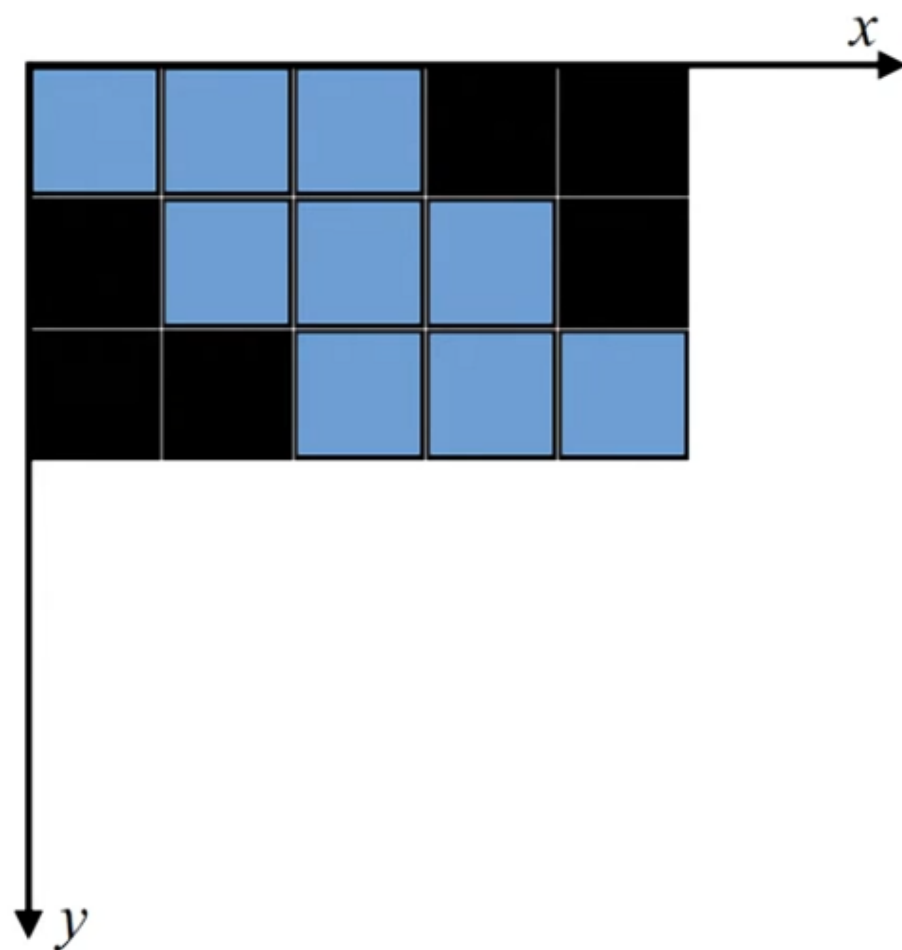
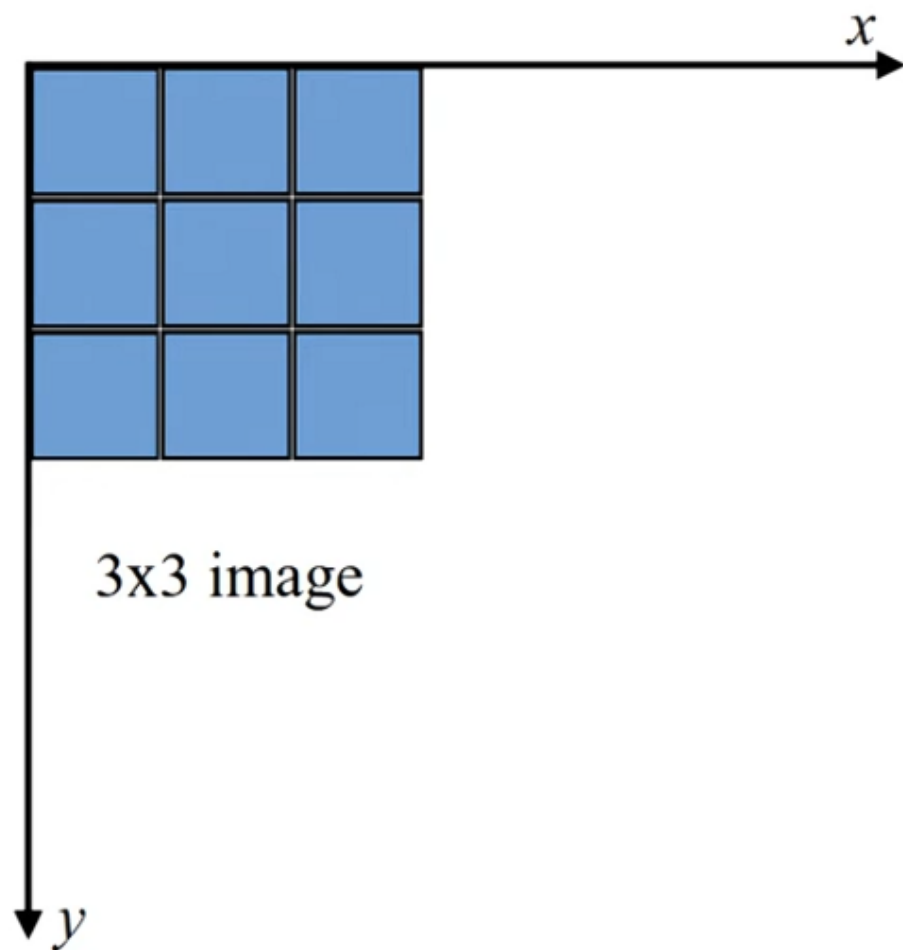
$$x' = x + \alpha y$$

$$y' = \beta x + y$$

α and β are shear parameters for horizontal and vertical directions respectively

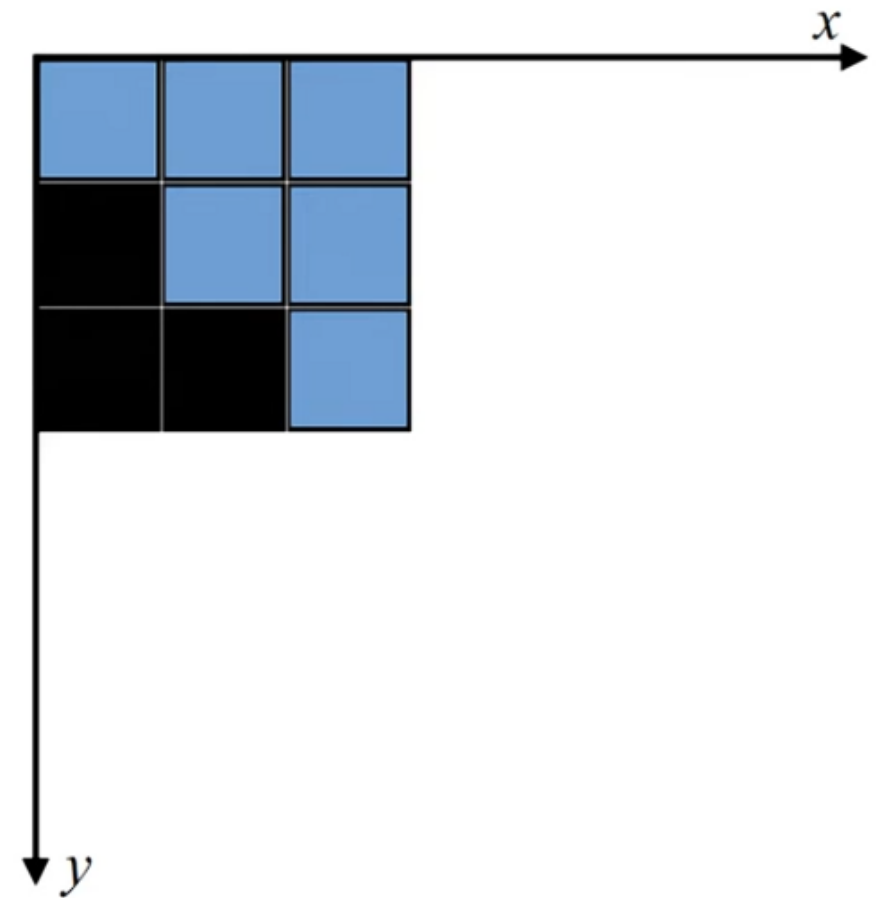
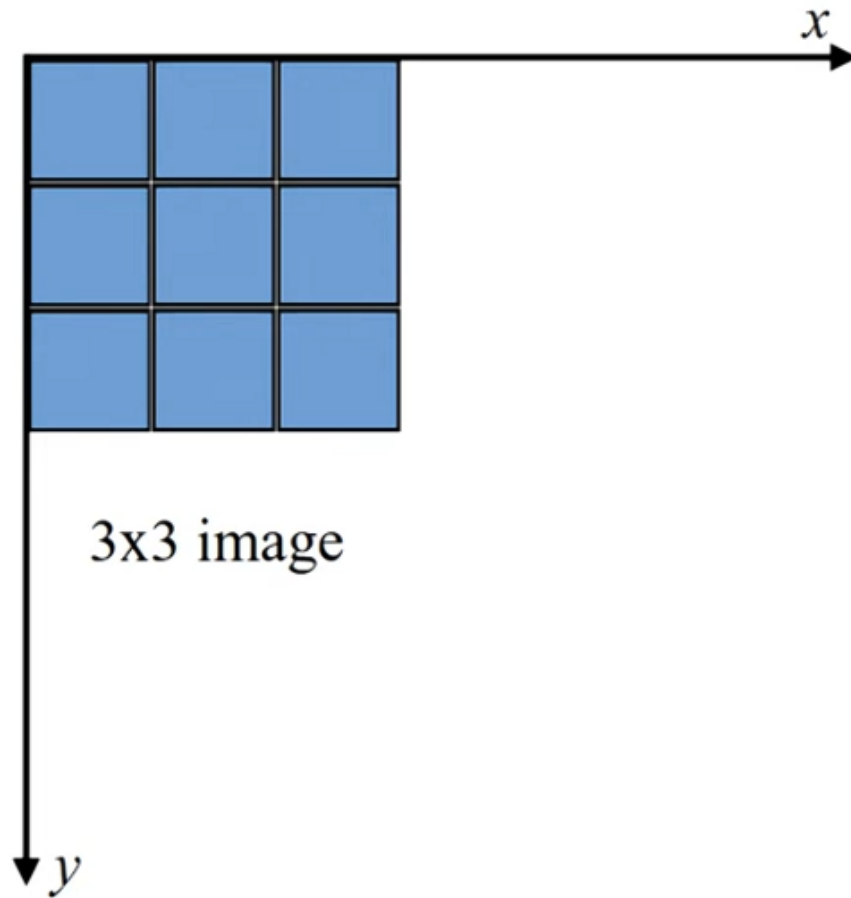
Horizontal shear with $\alpha=1$ can be applied
with this equation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



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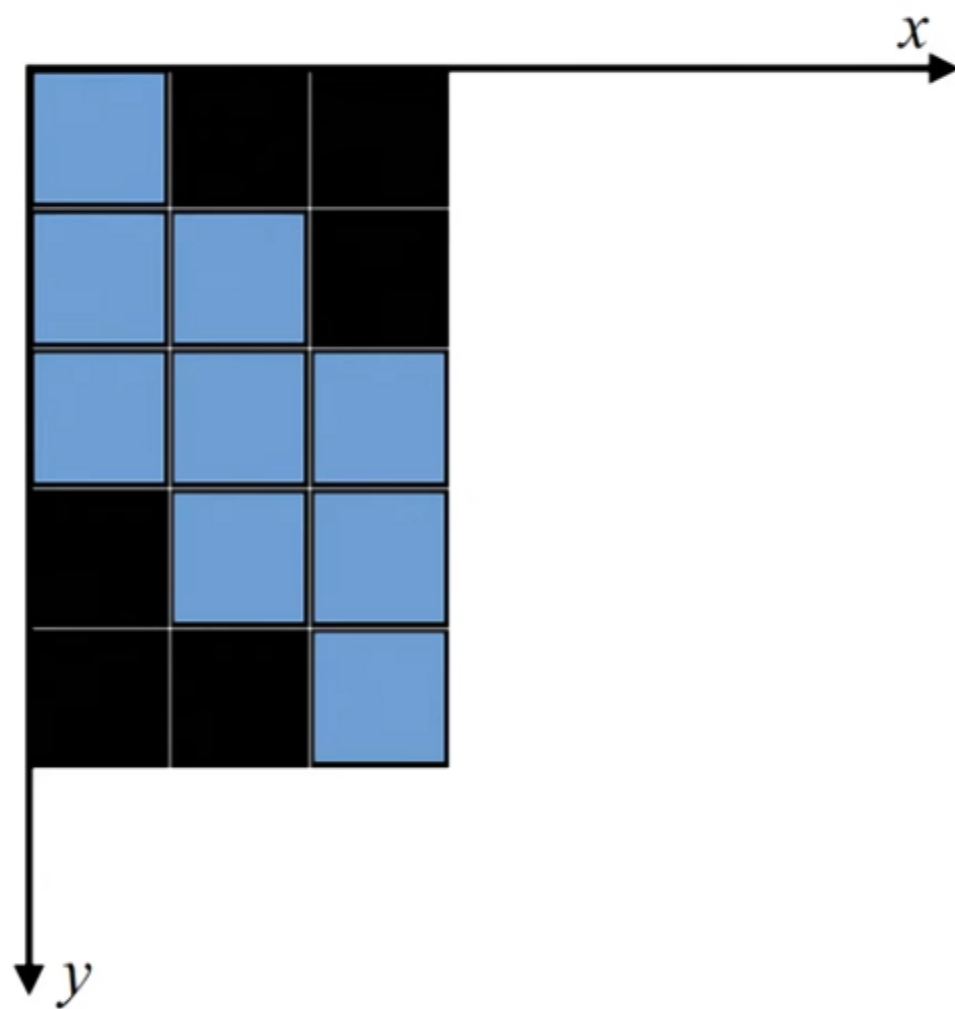
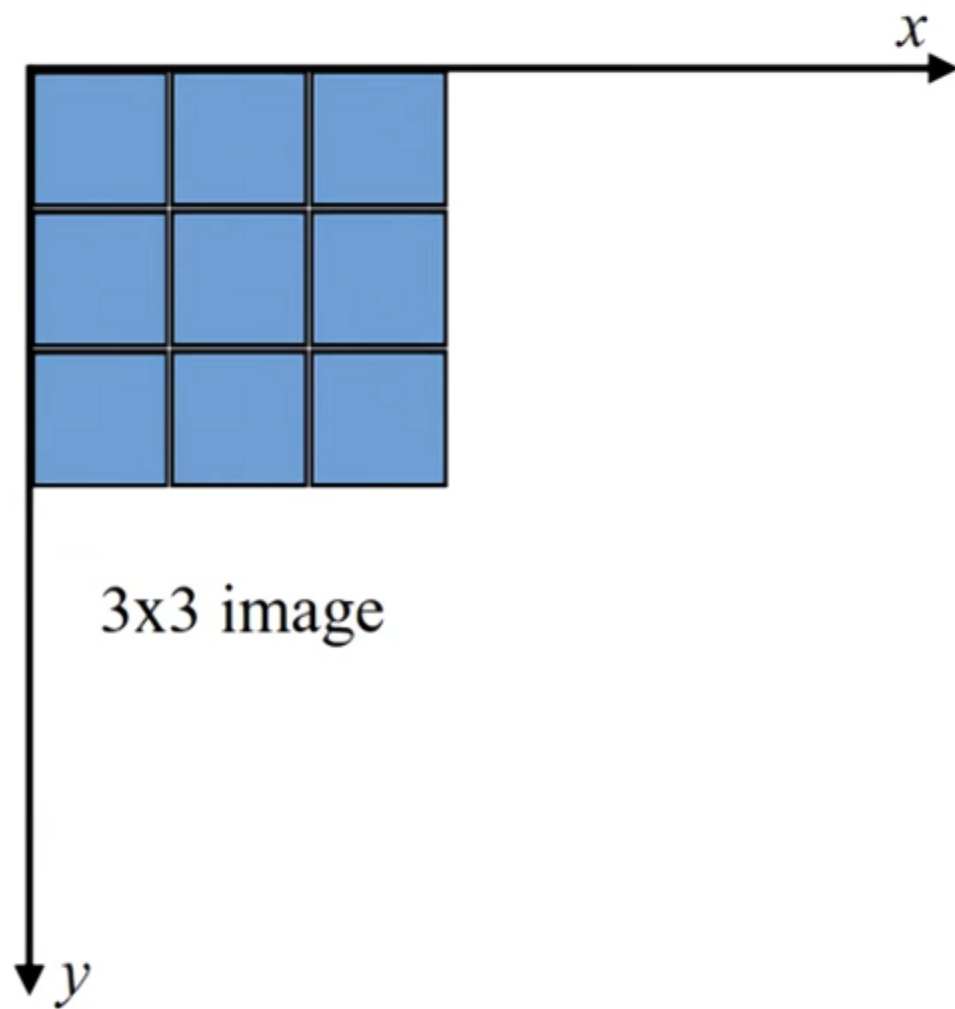
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



Pixels remaining out of original image
boundary can be cropped out

Vertical shear with $\beta=1$ can be applied
with this equation

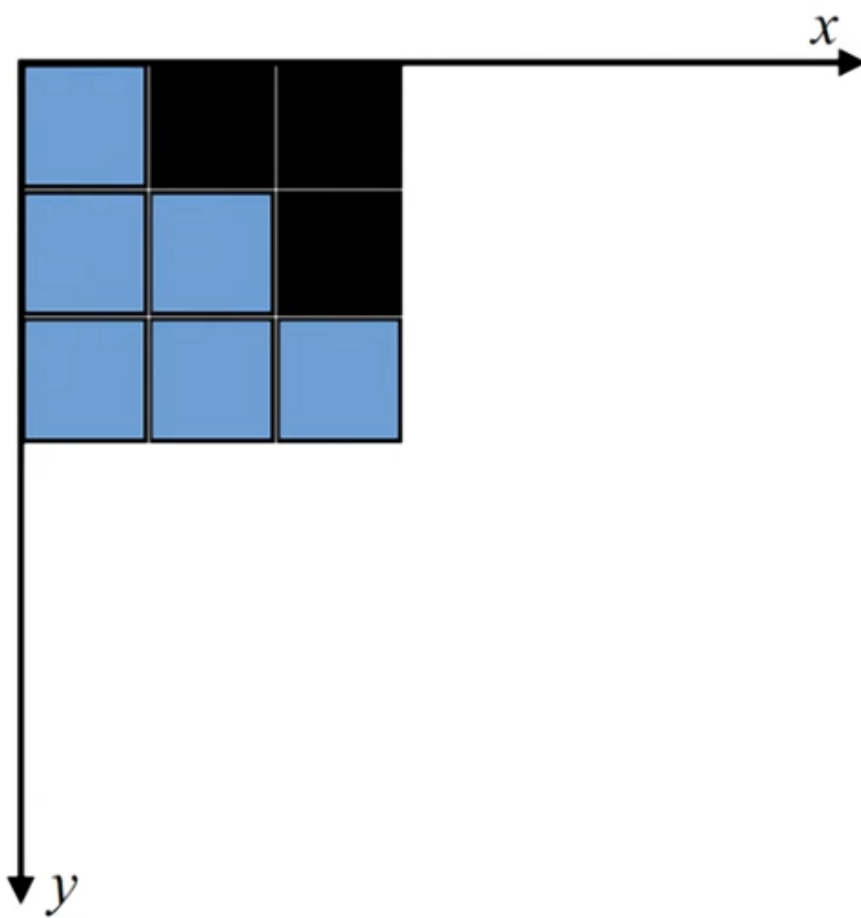
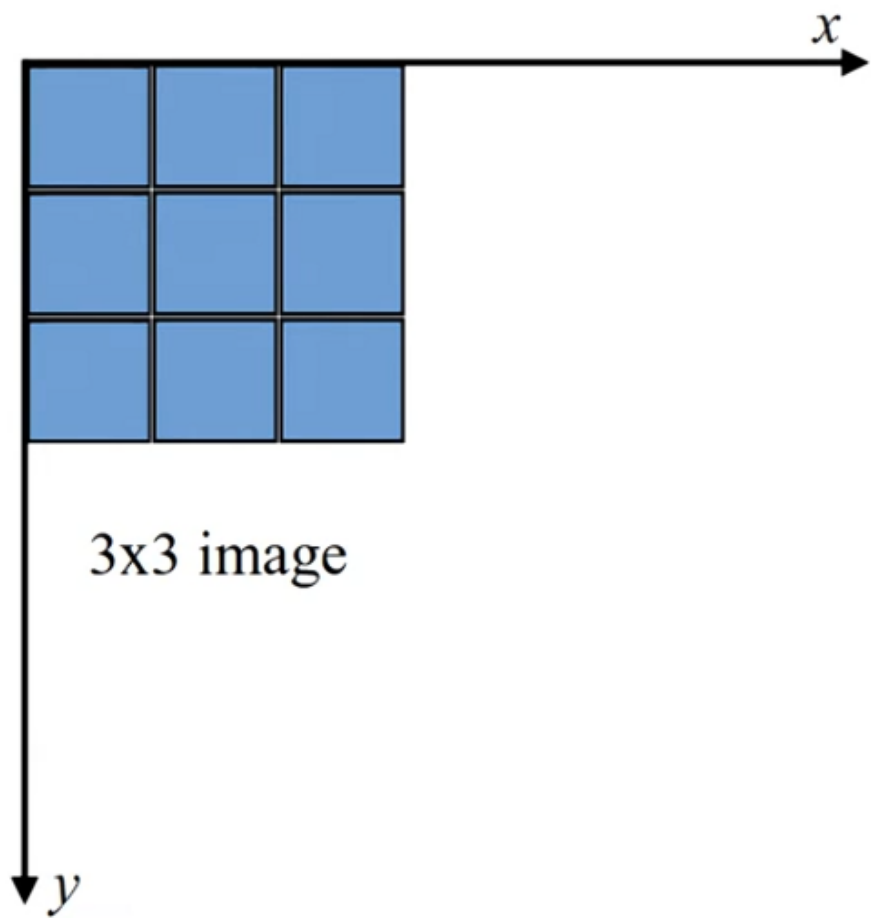
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$



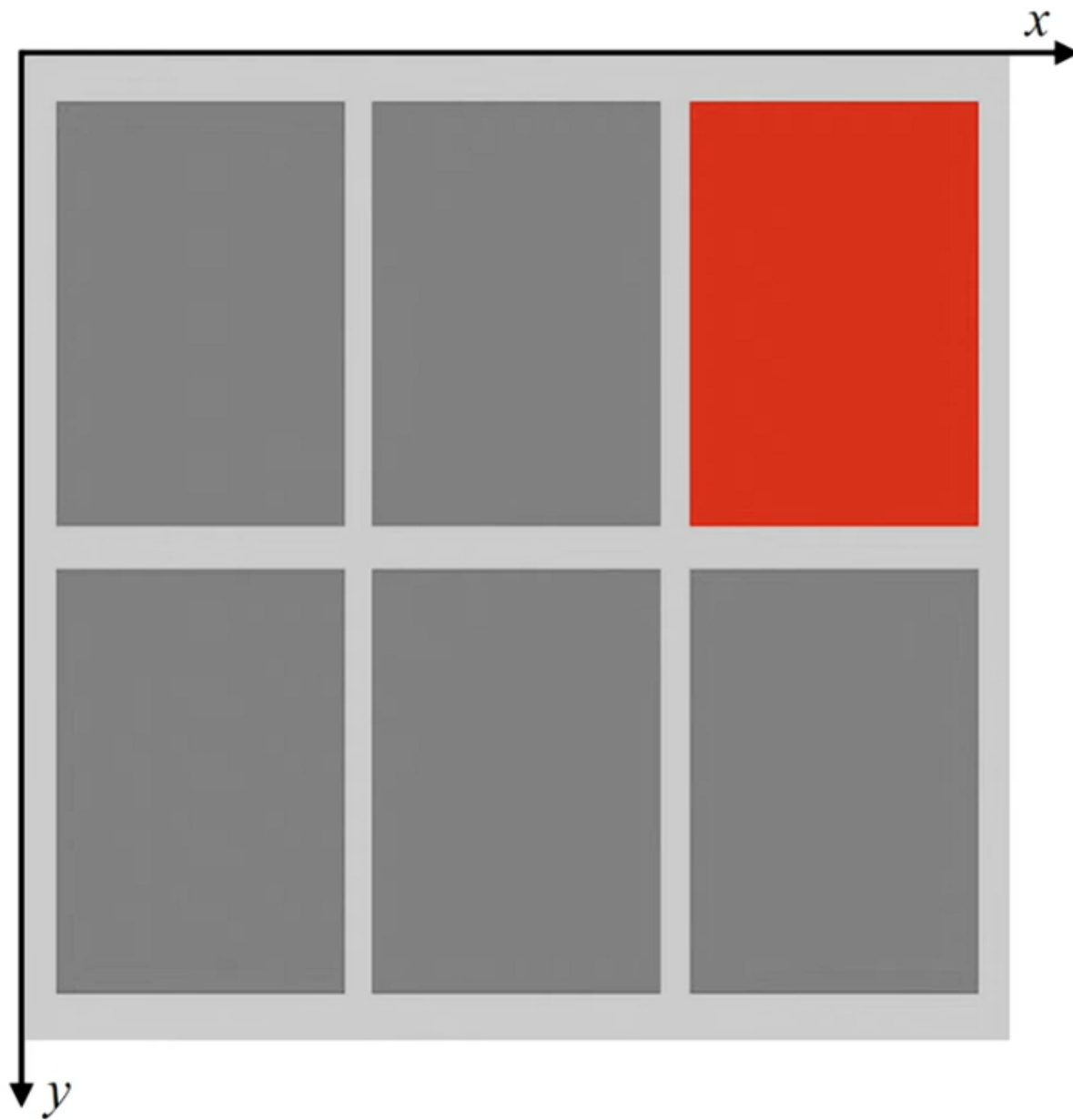
Pixels remaining out of original image
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Vertical shear with $\beta=1$ can be applied
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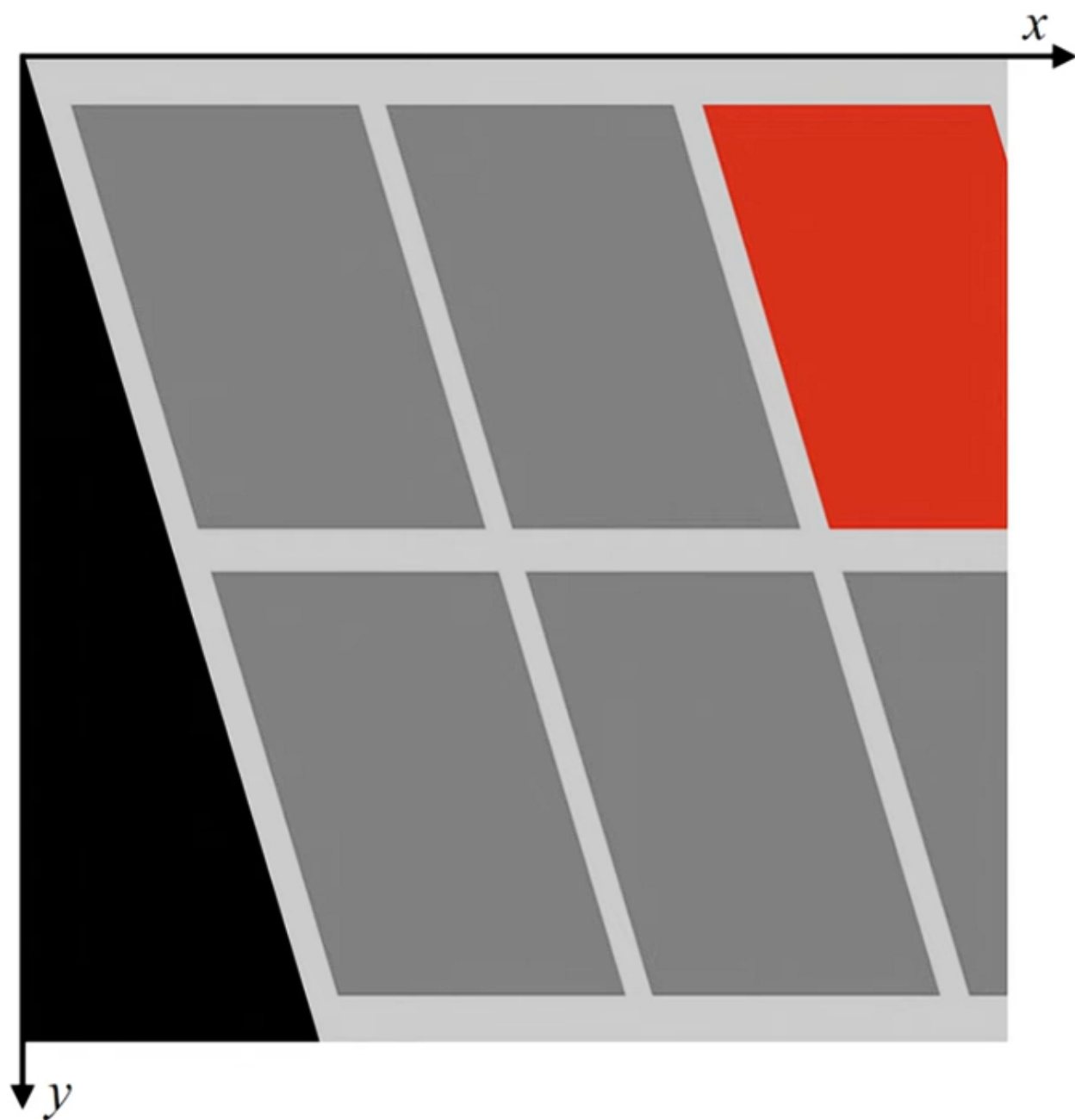


Pixels remaining out of original image
boundary can be cropped out



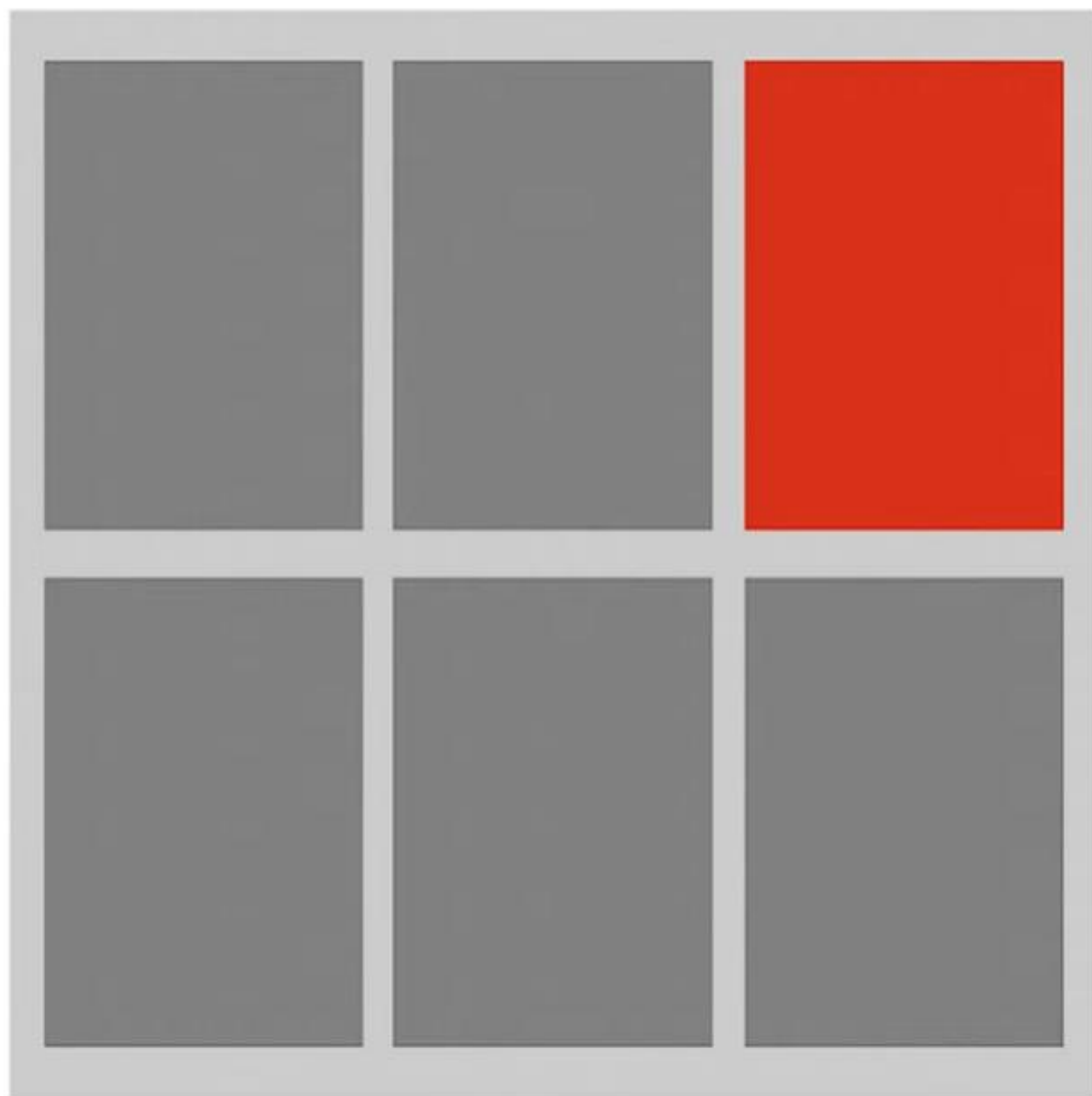
Matrix to apply horizontal
shear with $\alpha=0.3$

$$\begin{bmatrix} 1 & 0.3 \\ 0 & 1 \end{bmatrix}$$



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shear with $\alpha=0.3$

$$\begin{bmatrix} 1 & 0.3 \\ 0 & 1 \end{bmatrix}$$



Original Image

$\alpha=0.3$

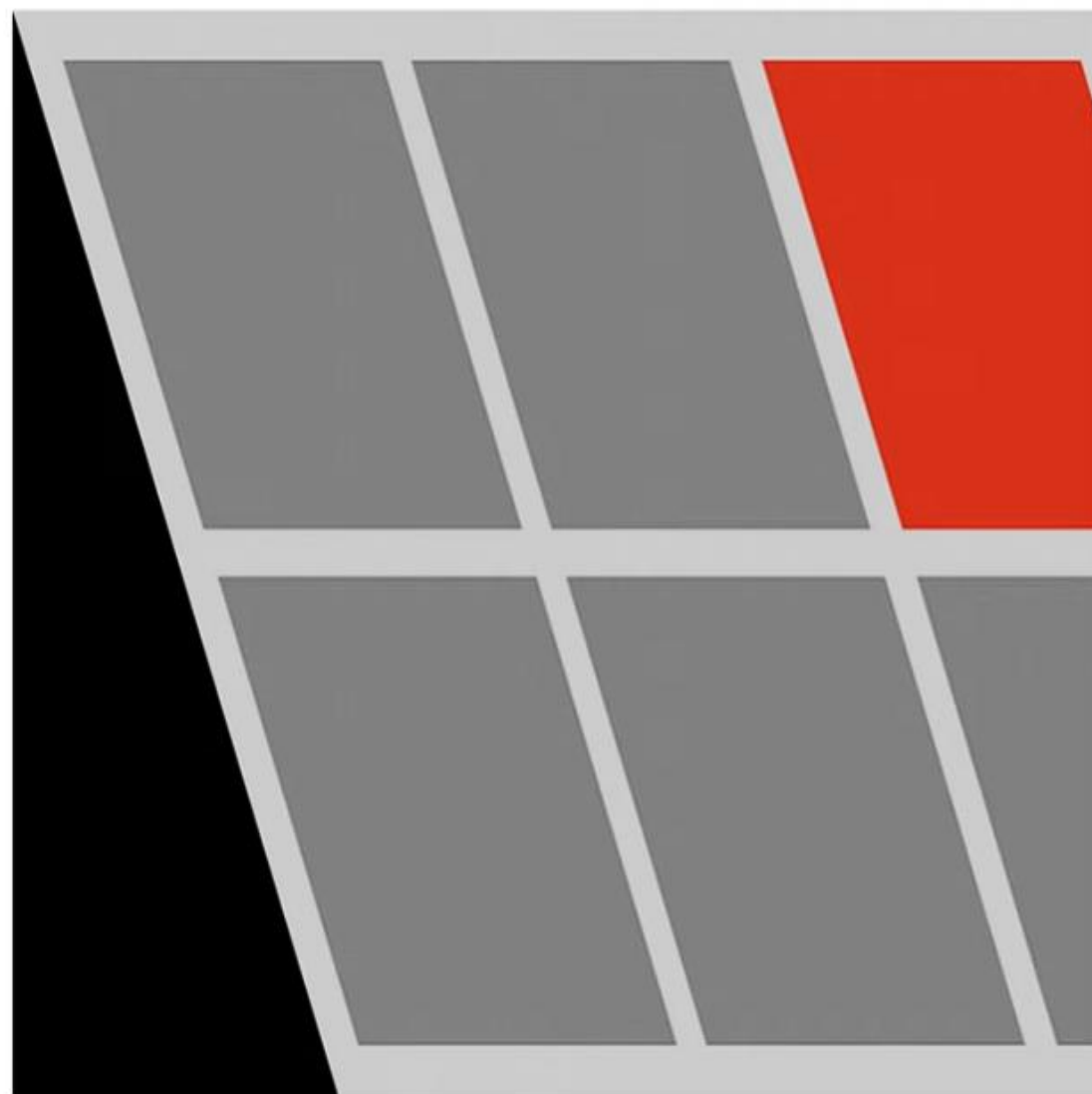
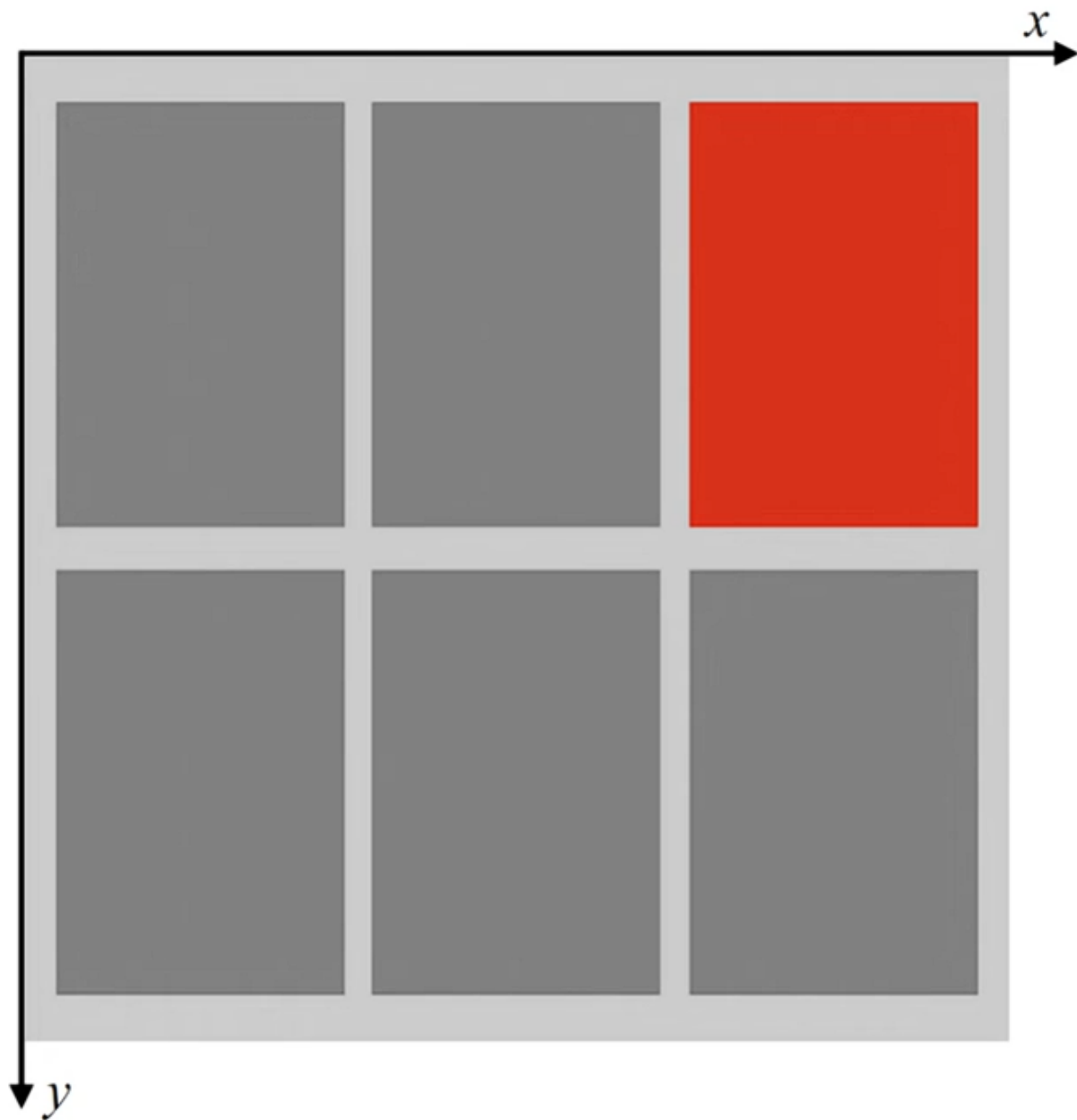
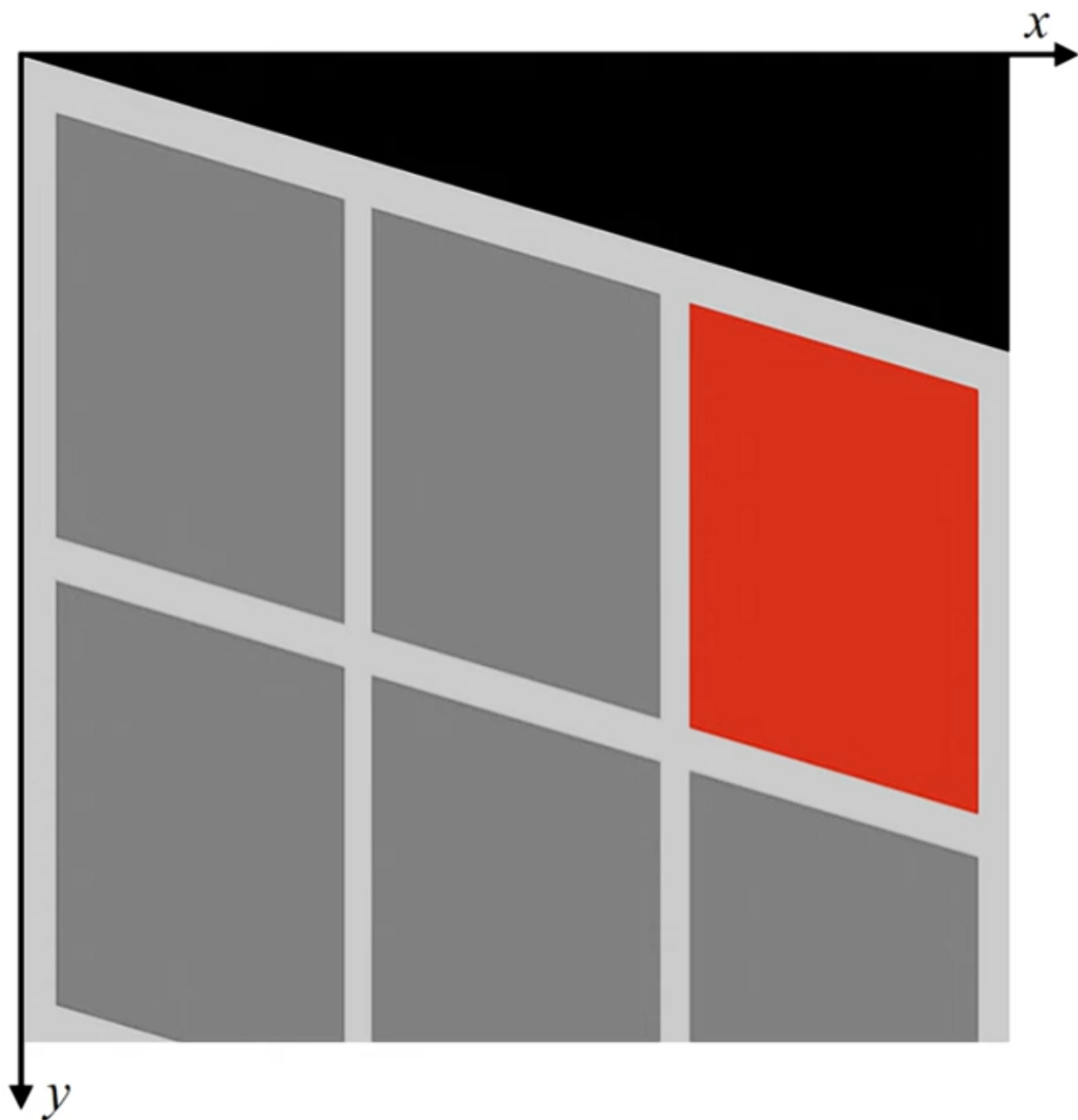


Image with Horizontal Shear



Matrix to apply vertical
shear with $\beta=0.3$

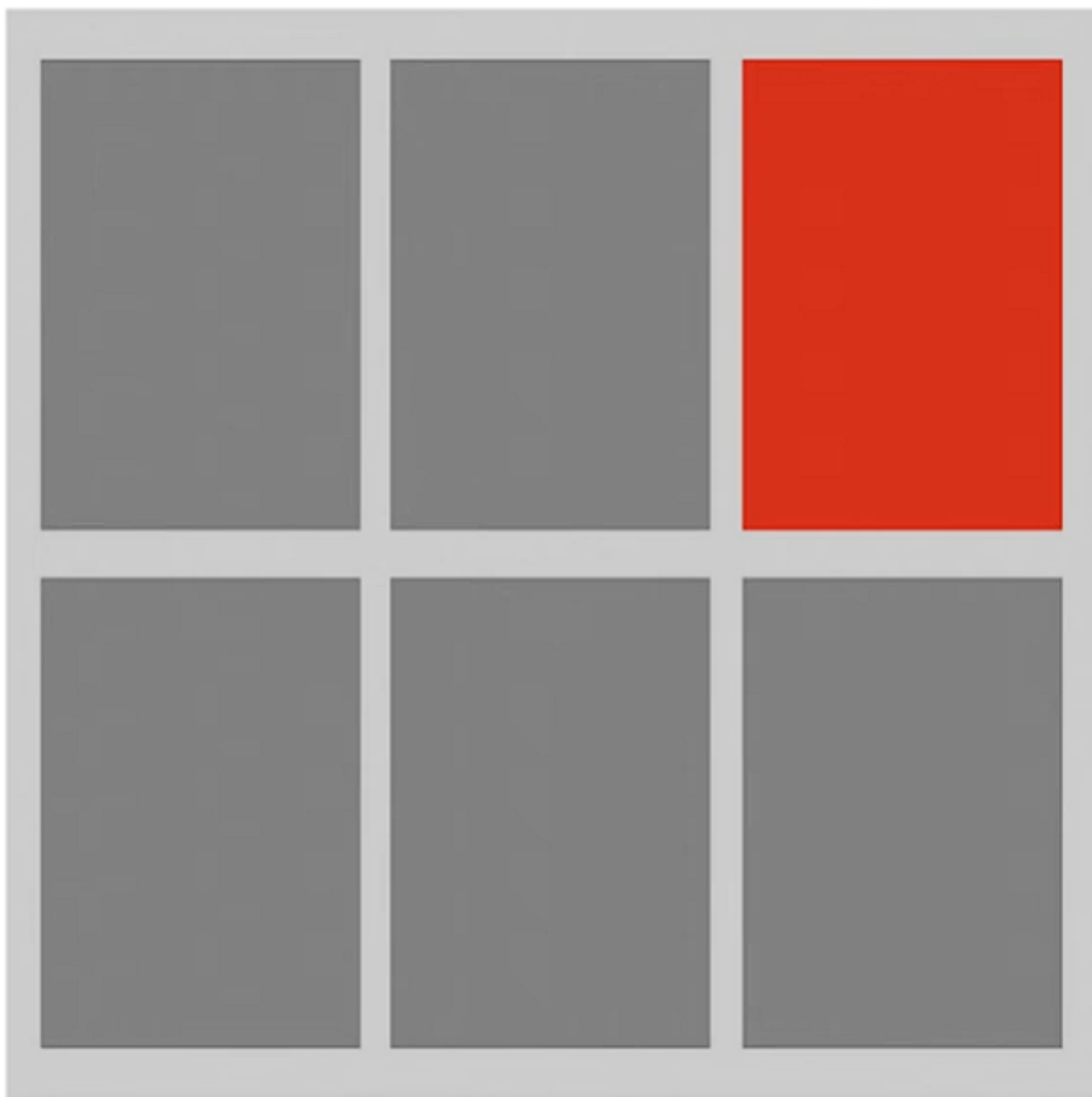
$$\begin{bmatrix} 1 & 0 \\ 0.3 & 1 \end{bmatrix}$$



Matrix to apply vertical
shear with $\beta=0.3$

$$\begin{bmatrix} 1 & 0 \\ 0.3 & 1 \end{bmatrix}$$

$$\beta = 0.3$$



Original Image

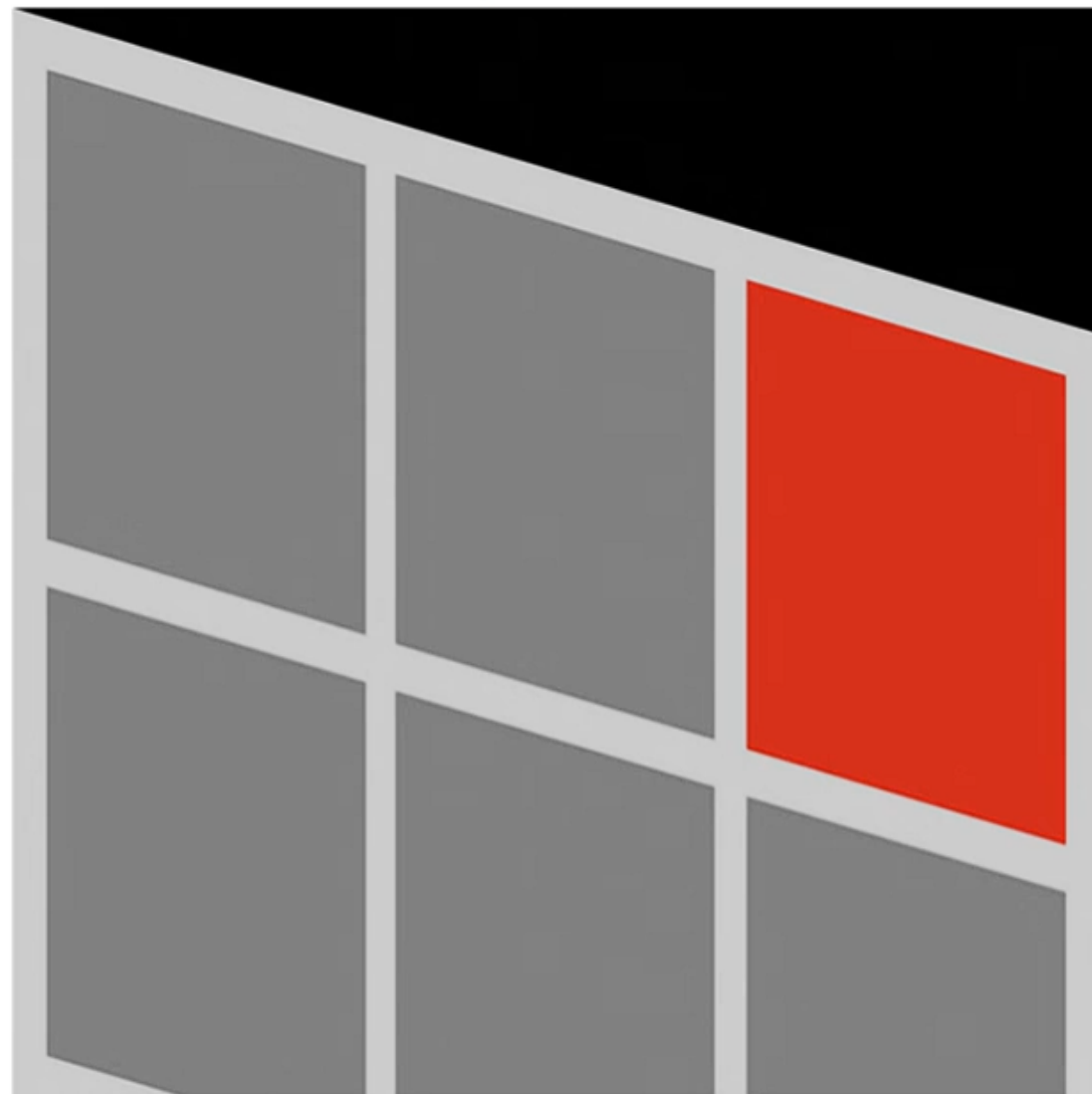
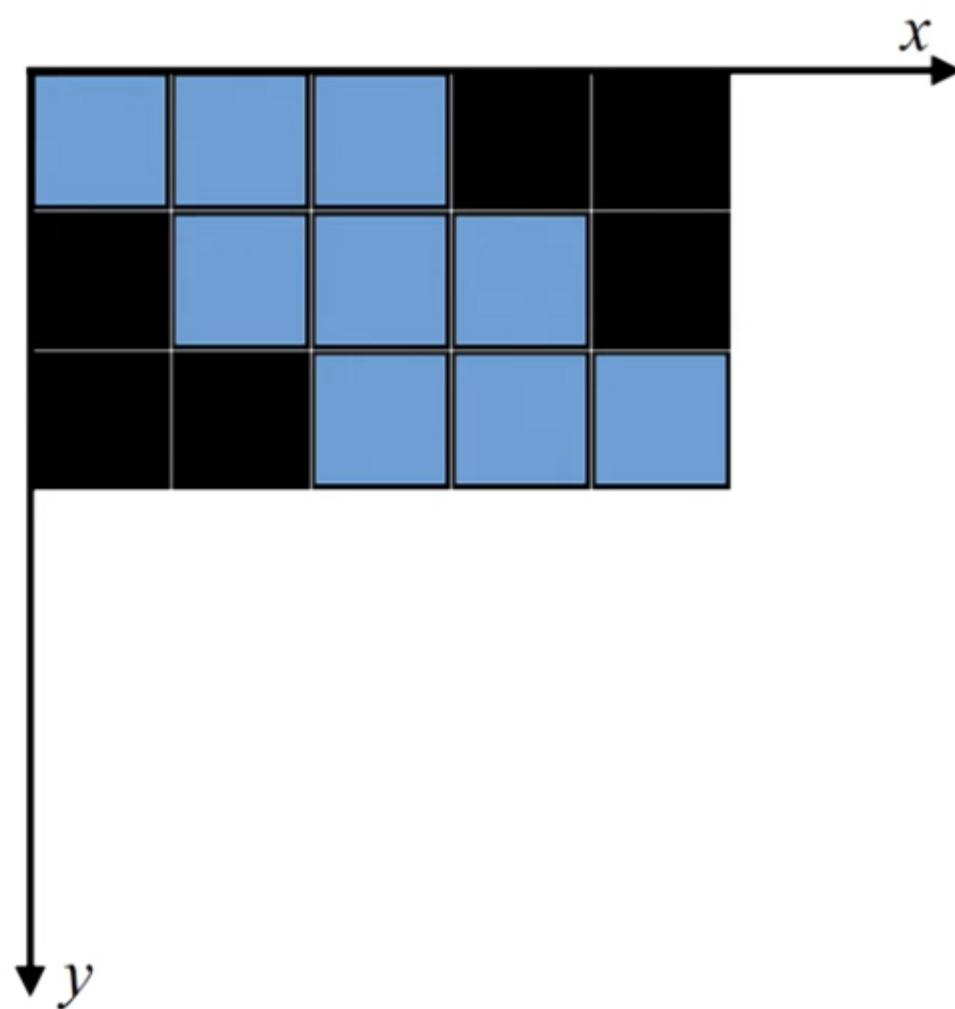
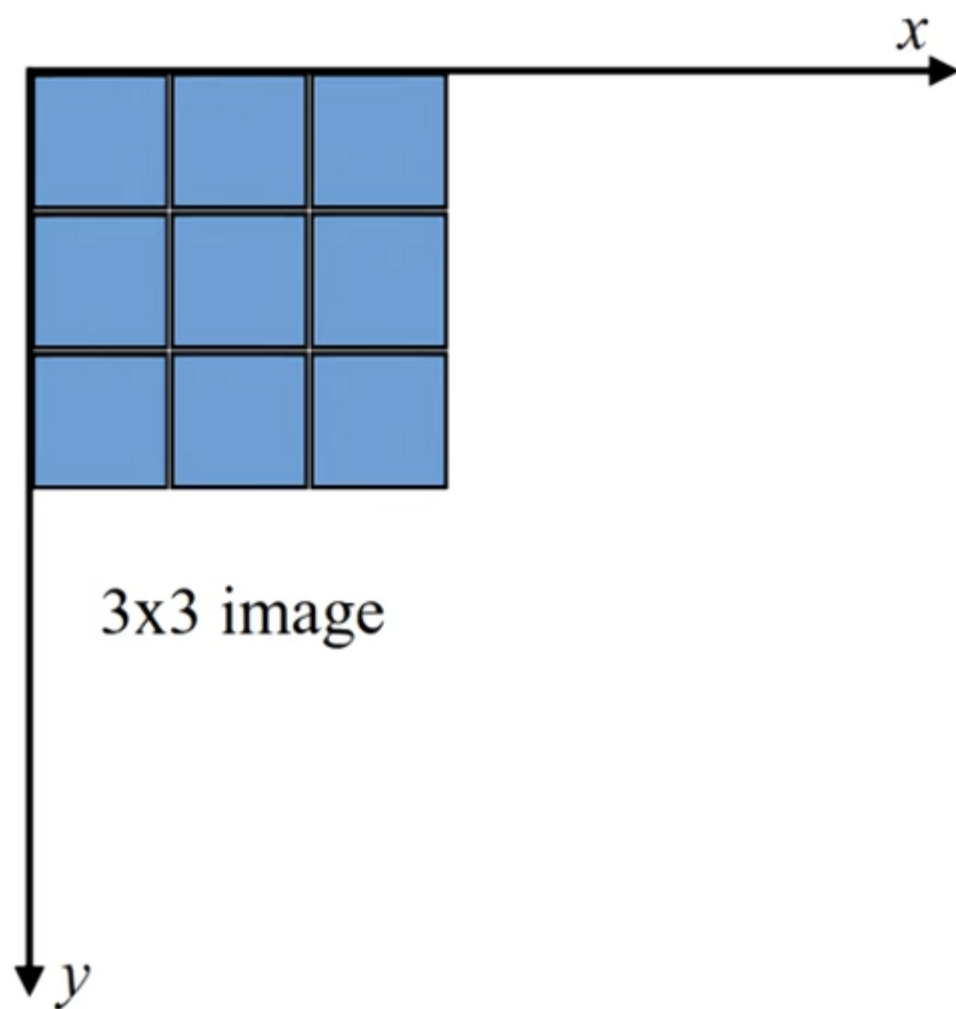
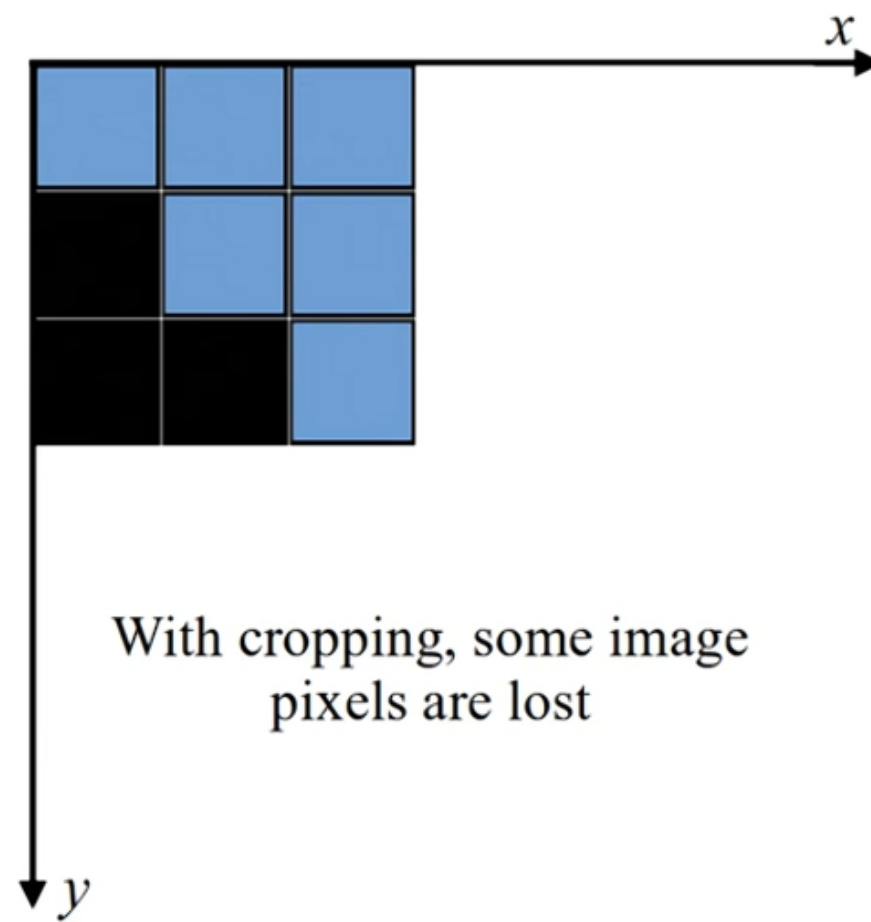
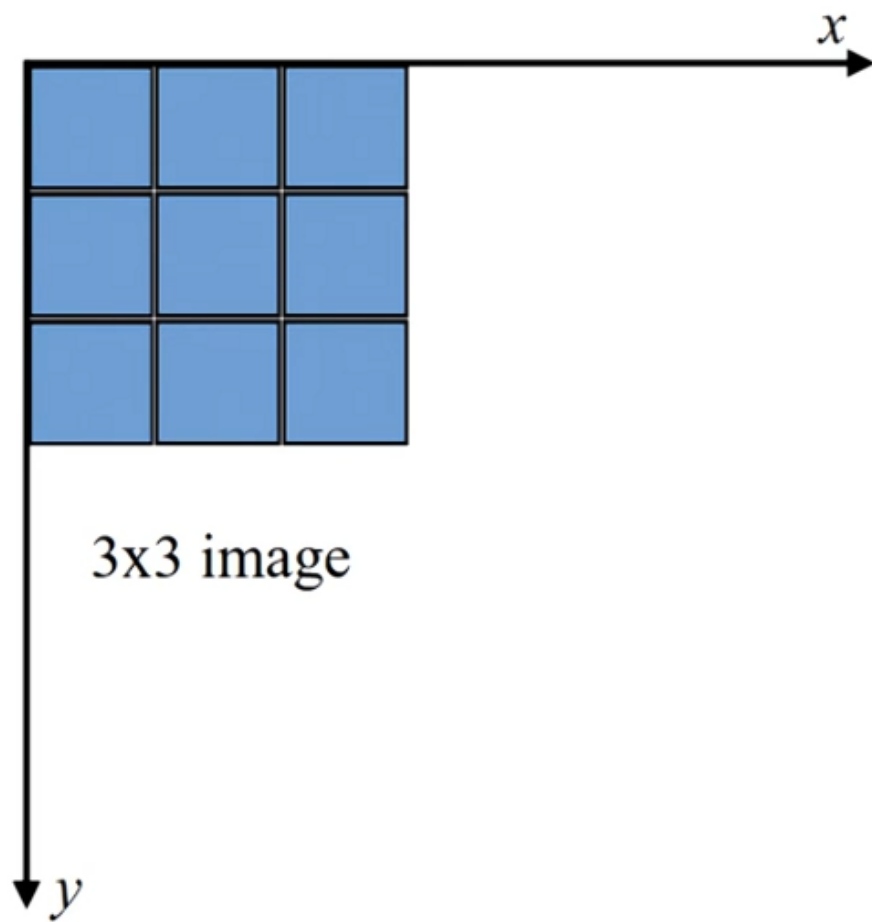


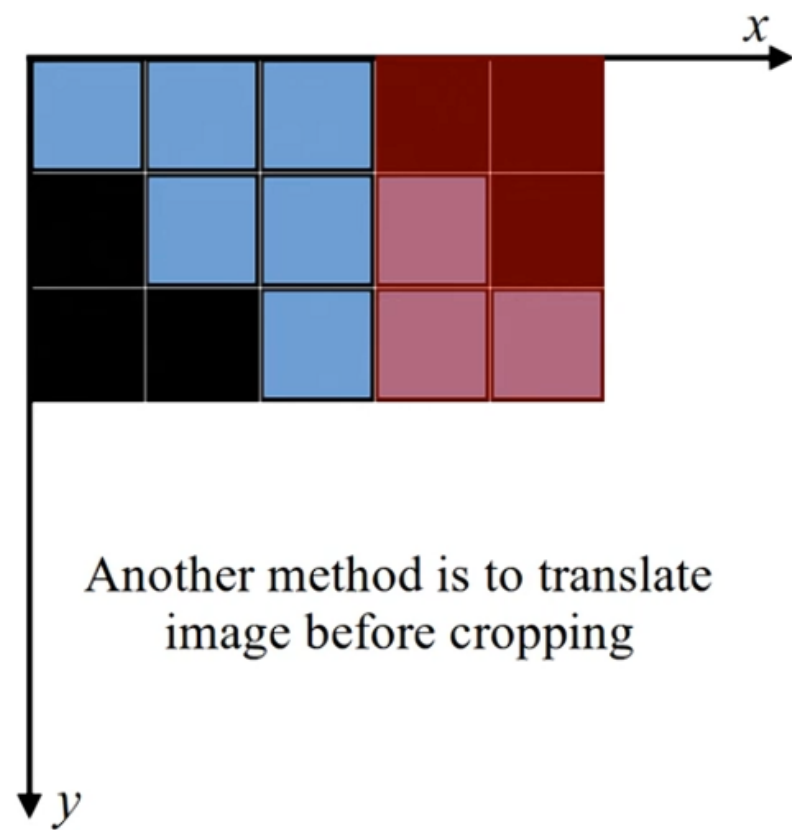
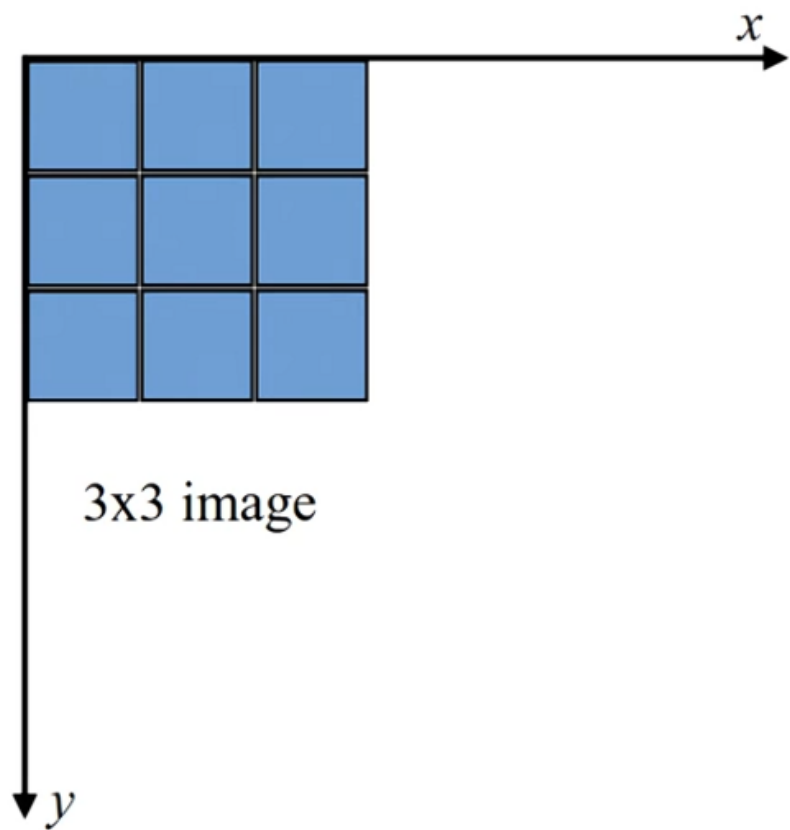
Image with Vertical Shear

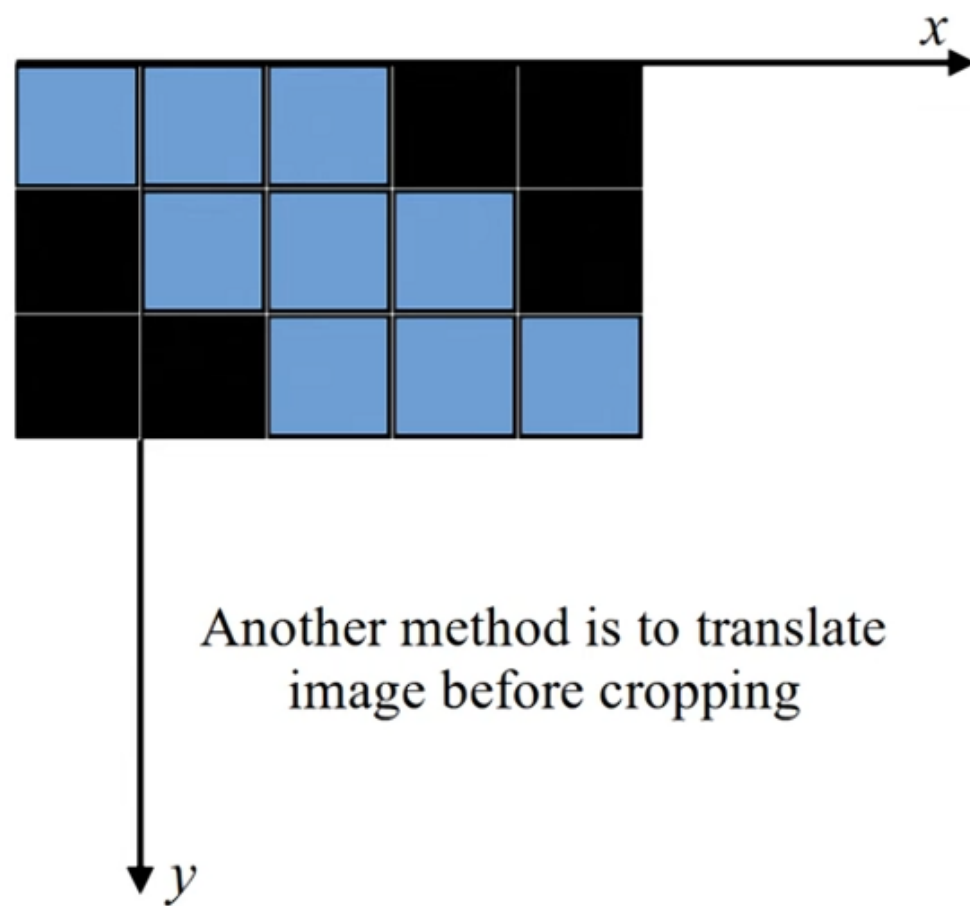
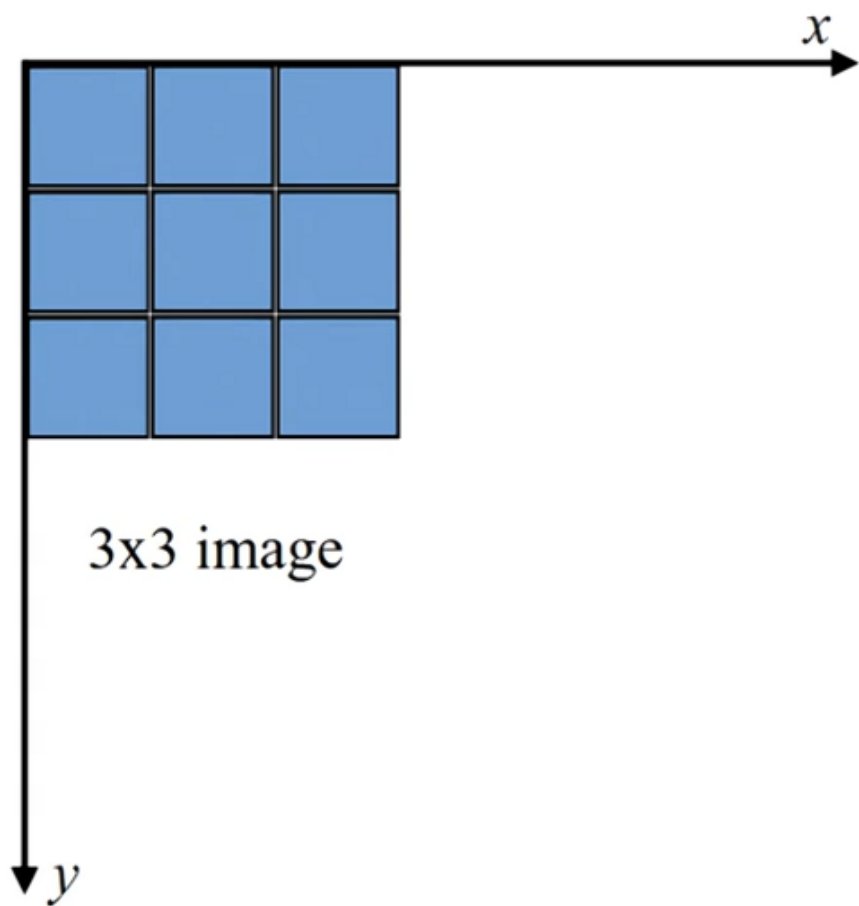
Shear with Affine Transformation

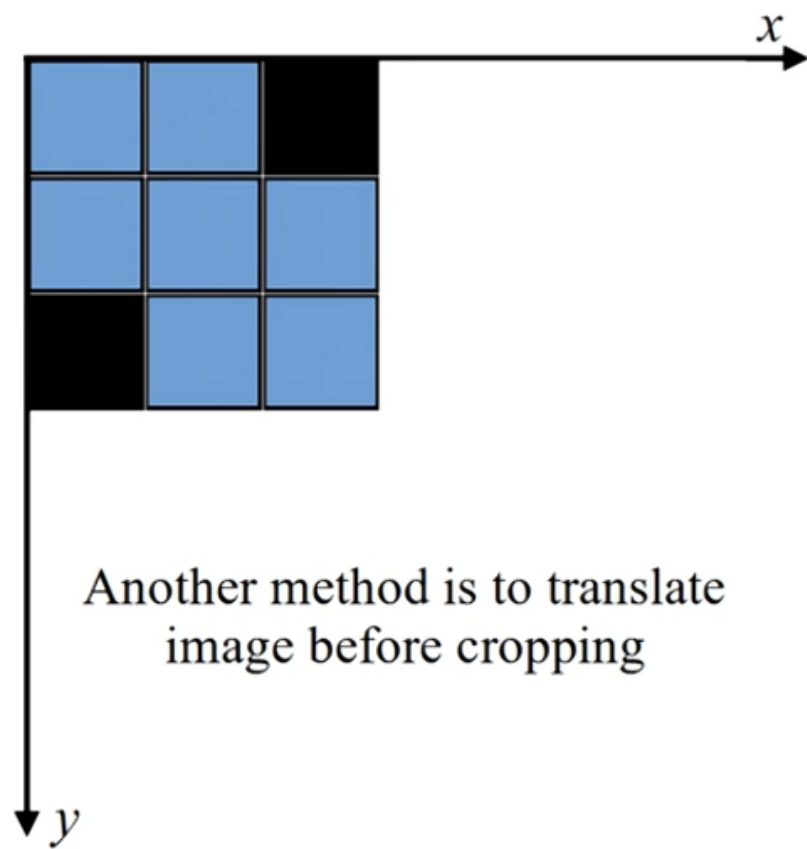
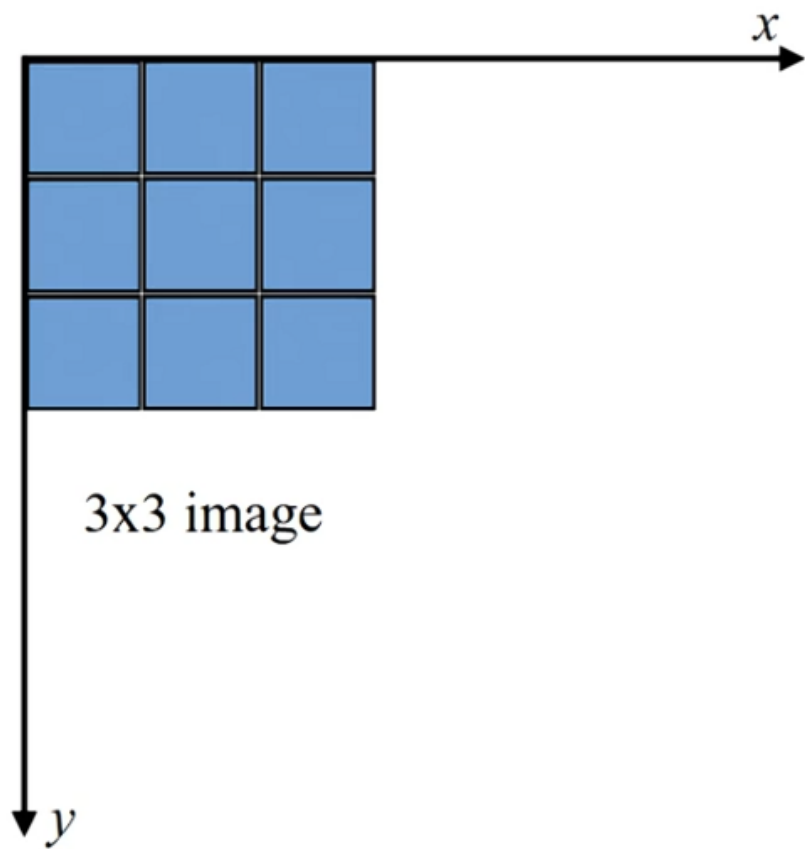
These are 3x3 image and its transformed version after
applying horizontal shear











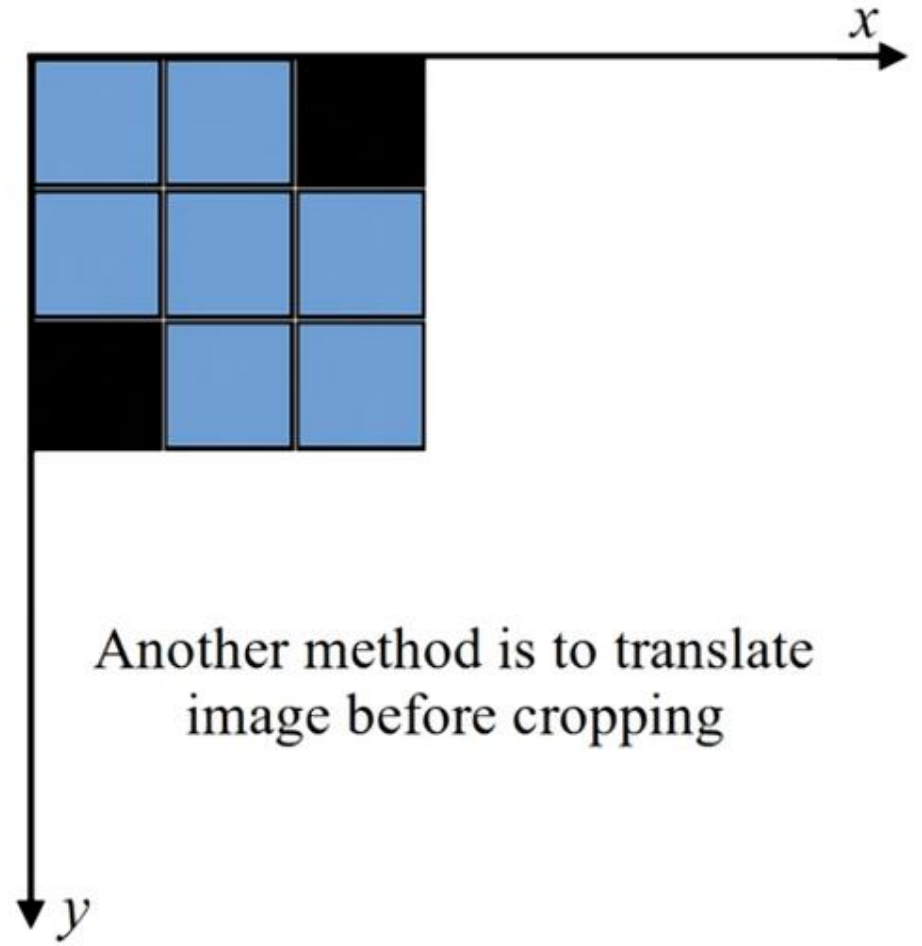
Translation and shear together form
an Affine Transformation

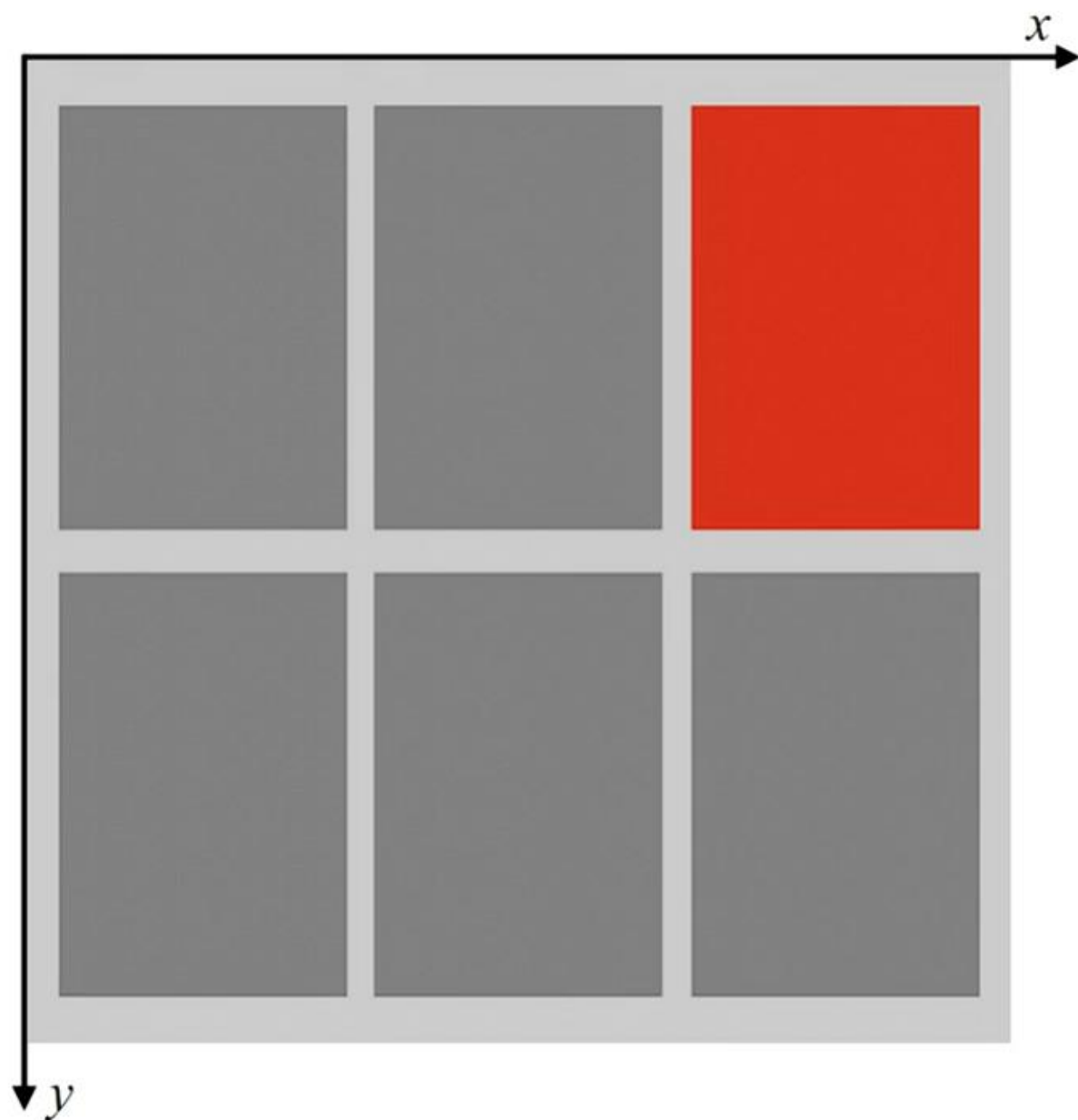
Cartesian
coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & \alpha \\ \beta & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

Homogeneous
coordinates

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha & b_0 \\ \beta & 1 & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$





Equation for horizontal shear by $\alpha=0.3$ with translation

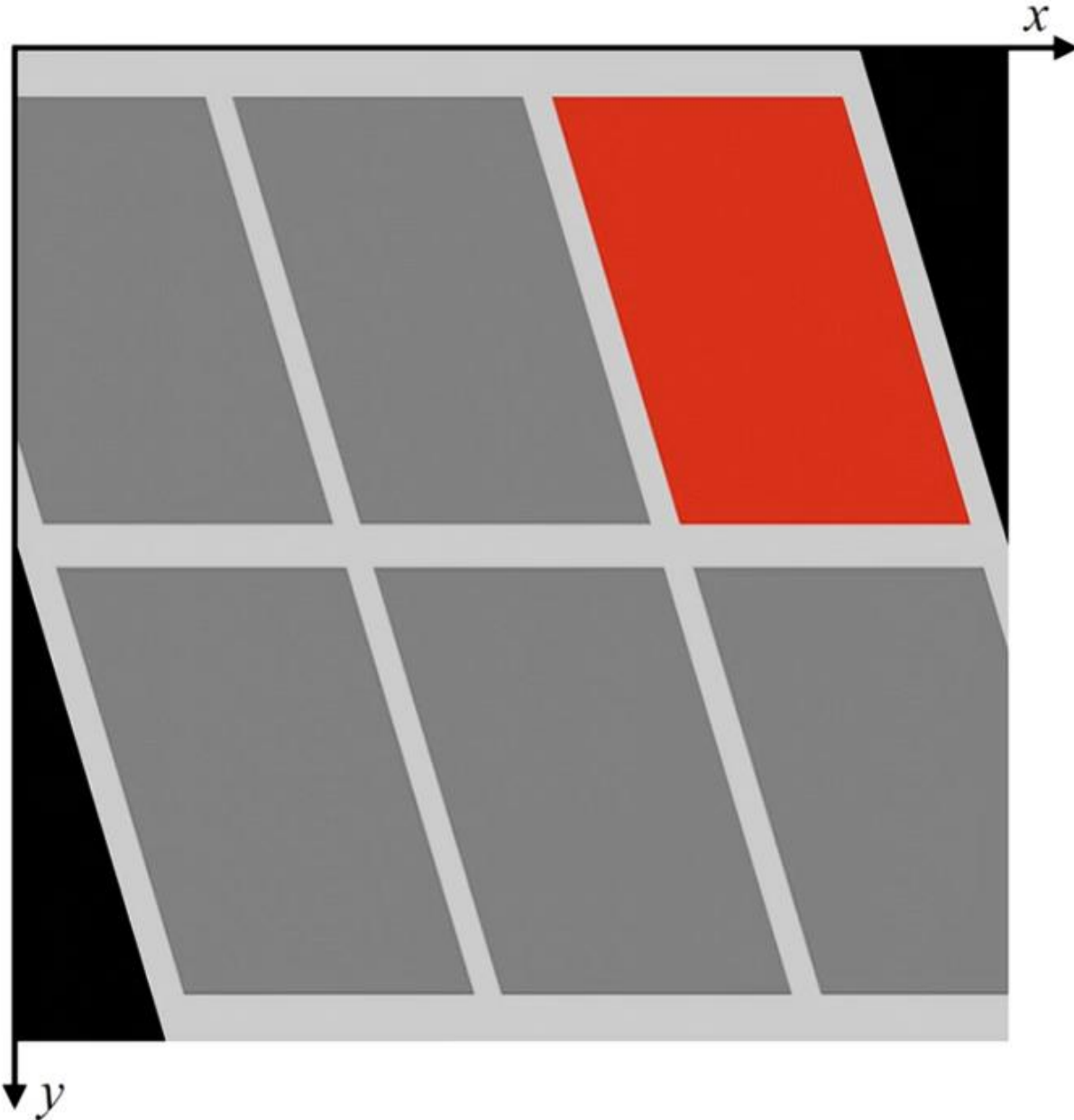
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha & b_0 \\ 0 & 1 & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

b_0 is negative of half of the max horizontal row shift due to shear

$$\text{max_row_shift} = \alpha \times \text{image_height}$$

$$b_0 = -0.5 \times \text{max_row_shift}$$

b_1 is zero, because shear is horizontal



Equation for horizontal shear by
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$$\alpha = 0.3$$

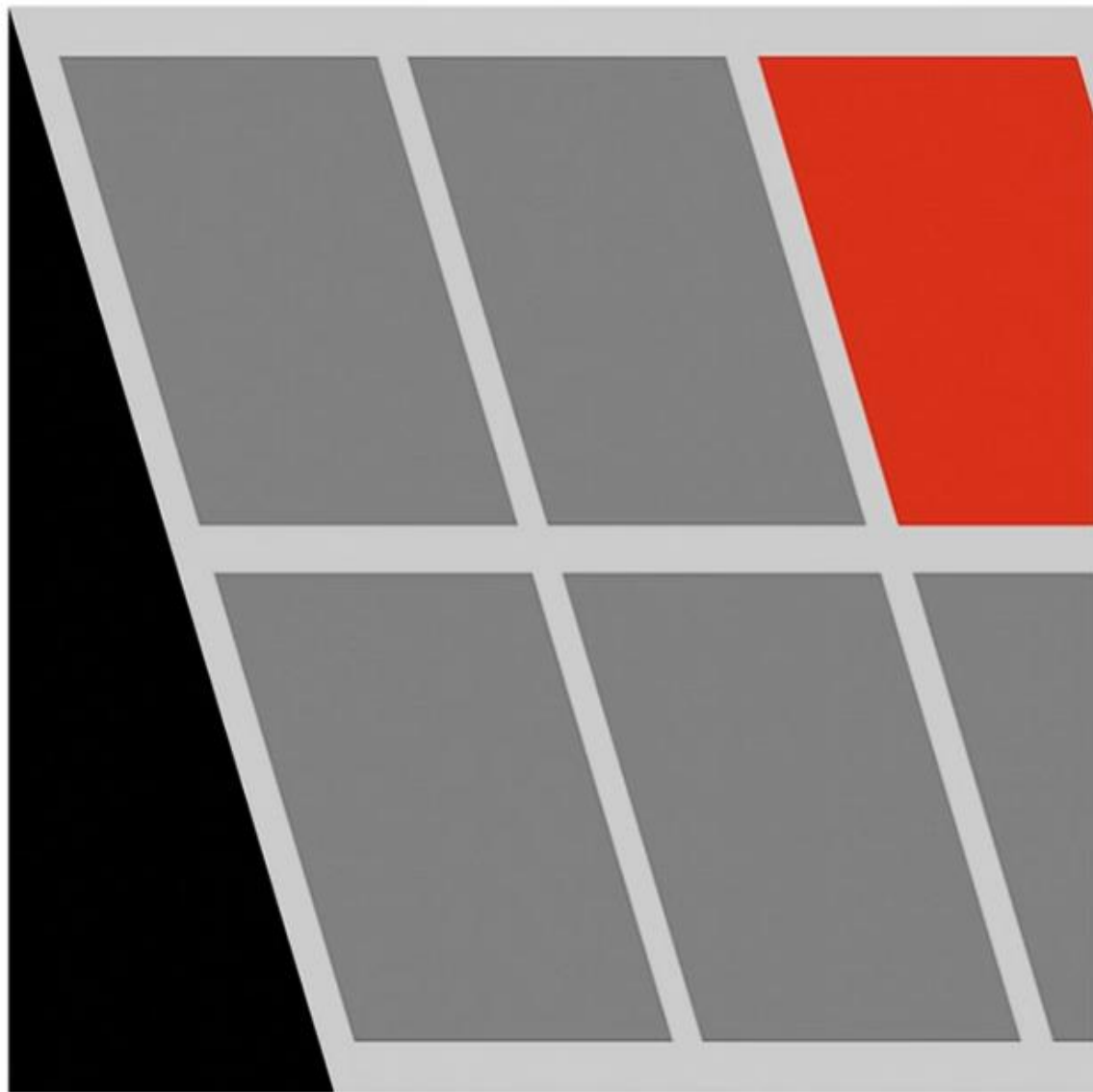


Image with Horizontal Shear

$$b_0 = -0.5 \times \text{max_row_shift}$$

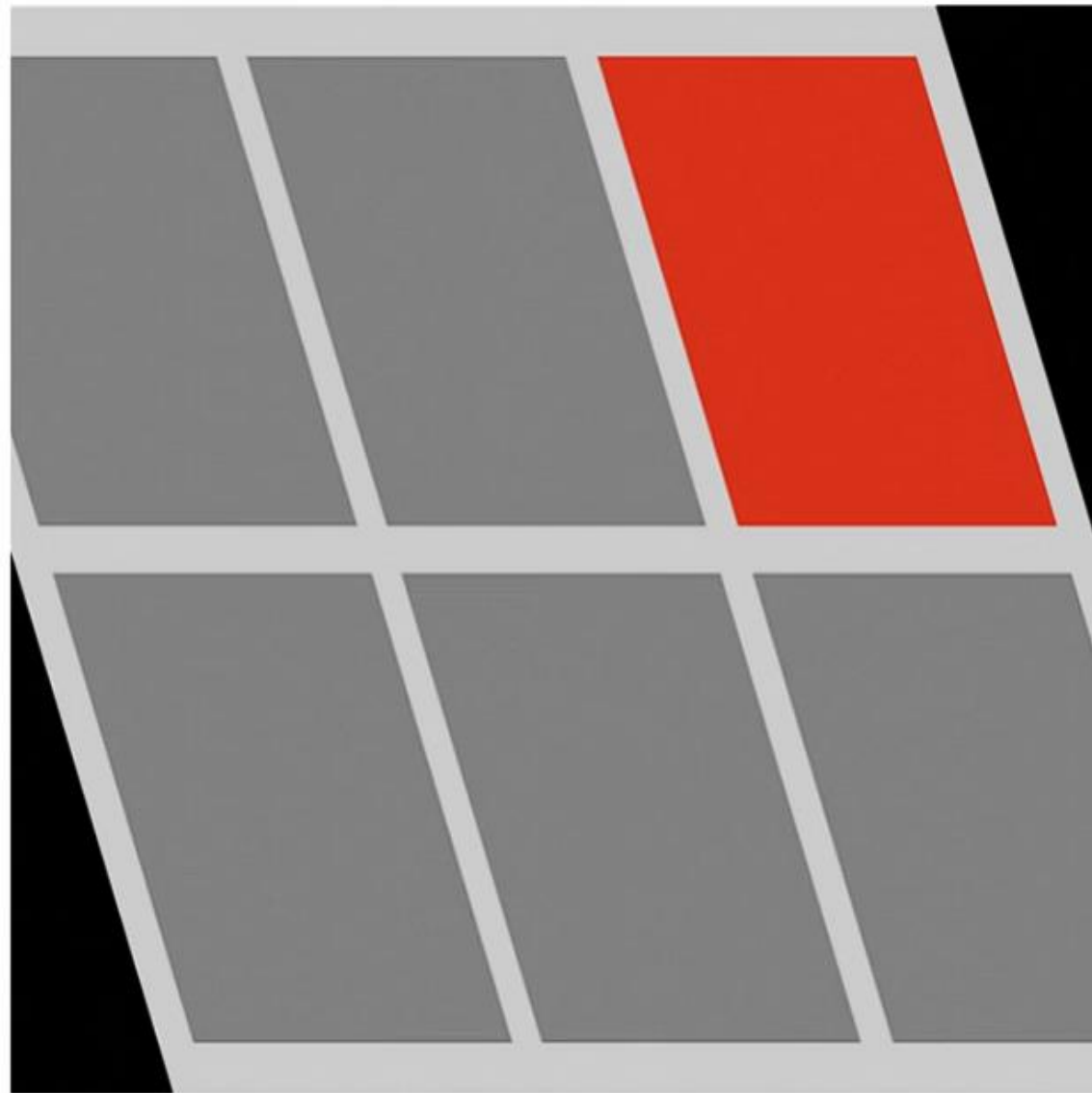


Image with Horizontal Shear and Translation

Reference

- https://www.youtube.com/watch?v=T_aGLwyoPrE&list=PLcEc12xiYZ7r24dJWP4GYwwXUgGS-wl4_&index=12